

Supreme Counselling for Personal Development

Infrastructure Proposal — v2

Cloudflare-Native Public + Azure Internal (Staff Company Data)

Cloudflare Pages + Workers + D1 + R2

Azure ACA + VNet + VPN

BBD \$0–40/mo

Public-facing (Cloudflare)
Pages · Workers · D1 · R2 · AI

**BBD \$140–200/
mo**

Internal (Azure)
ACA · Storage · VNet · VPN

Prepared: June 2026 • Region: US East

Staff access via Azure VPN Gateway (Azure AD auth — uses existing Microsoft domain)

1. Executive Summary

This proposal presents a **two-part infrastructure** for Supreme Counselling for Personal Development:

- **Public-facing** — website, bookings, AI chat — runs entirely on **Cloudflare's edge platform** at near-zero cost
- **Internal staff system** — company data RAG — runs on **Azure** inside an isolated VNet, accessed via **Azure P2S VPN** using the company's existing Microsoft domain for authentication

Public and internal data are kept strictly separate — no shared databases, no complex RBAC. The public side costs **~BBD \$0–40/mo**. The internal side costs **~BBD \$140–200/mo**. Total: **~BBD \$140–240/mo**.

Key design principles: Separation of concerns • Data sovereignty • Zero fixed server costs for public side • Microsoft AD integration for staff access • No over-engineering

2. Architecture Overview

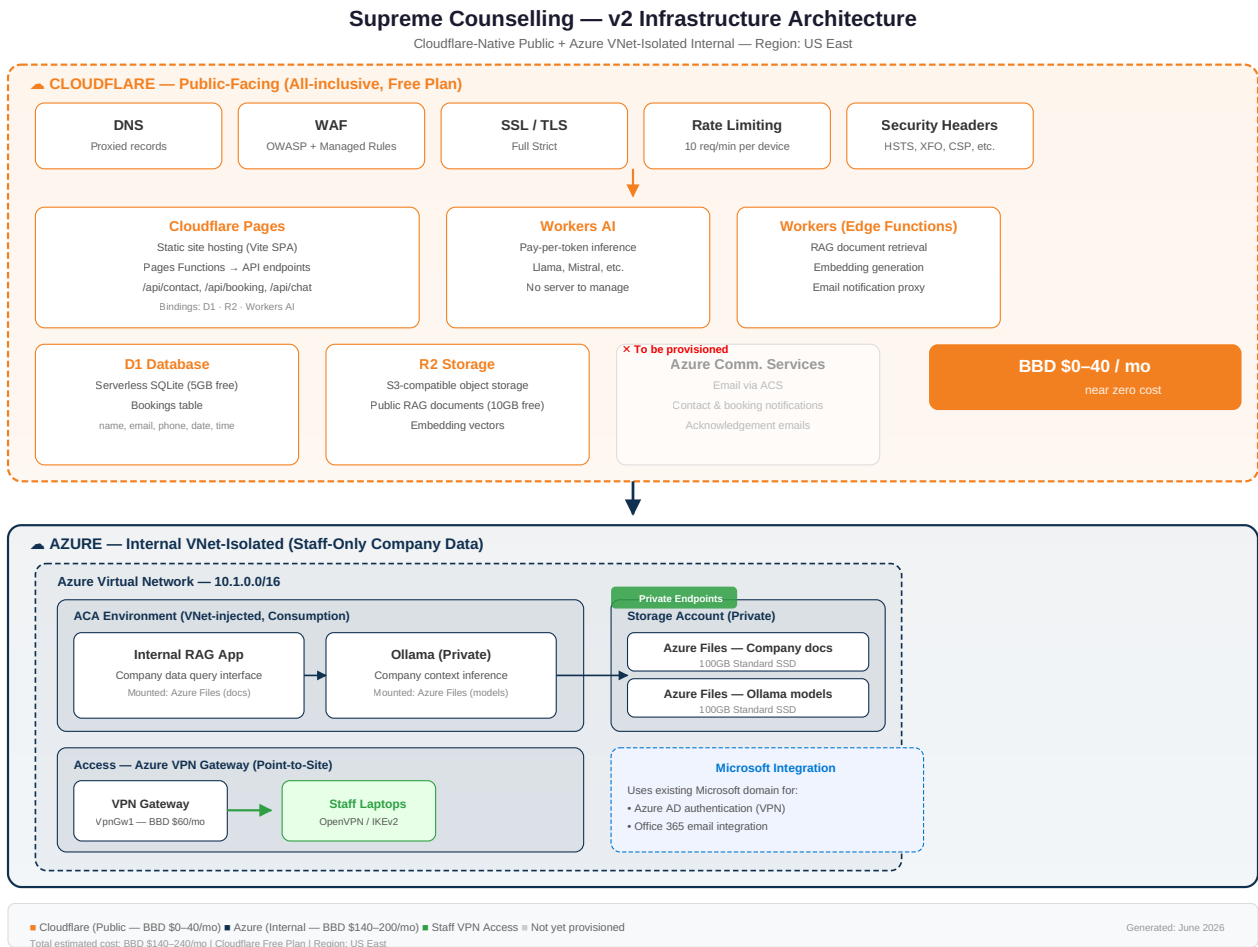


Figure 1: v2 Architecture — Cloudflare-native public + Azure VNet-isolated internal

Part 1 — Cloudflare (Public-Facing, Free Plan)

- **Cloudflare Pages** — Vite SPA static hosting (unlimited requests)
- **Pages Functions** — API endpoints for /api/contact, /api/booking, /api/chat
- **D1 Database** — serverless SQLite for bookings
- **R2 Storage** — public RAG documents + embedding vectors
- **Workers AI** — chat inference (pay-per-token, no server)
- **Workers** — RAG retrieval, email proxy
- **Cloudflare DNS, WAF, SSL, Rate Limiting** — edge security

Part 2 — Azure Internal (Staff-Only, VNet-Isolated)

- **Azure Virtual Network (10.1.0.0/16)** — isolated from internet
- **ACA Environment (VNet-injected)** — internal containers
- **Internal RAG App** — company data query interface
- **Ollama (Private)** — private AI inference on company data

- **Storage Account + Private Endpoints** — company docs + embeddings
- **Azure Files** — 100GB for company docs, 100GB for Ollama models
- **Azure VPN Gateway (P2S)** — staff access using Azure AD (Microsoft domain auth)

3. Cost Breakdown

All prices in **Barbadian Dollars (BBD)** at the pegged rate: 1 USD = 2 BBD. Based on US East pricing.

Cloudflare (Public)

Service	BBD/mo
Cloudflare Pages	\$0
Cloudflare Workers	\$0
Cloudflare D1 (5GB)	\$0
Cloudflare R2 (10GB)	\$0
Workers AI (pay-per-token)	~\$0–40
DNS / WAF / SSL	\$0
Rate Limiting	\$0
Subtotal	\$0–40

Workers AI: ~\$0.001–0.003 per query. Light usage stays under \$5–10/mo.

Azure (Internal)

Resource	BBD/mo
ACA — Internal RAG	~\$30
ACA — Ollama (Private)	~\$60
ACA Environment (VNet-injected)	~\$20
Azure Files (Docs + Models, 200GB)	~\$48
Blob Storage (Embeddings)	~\$4
Azure VPN Gateway (VpnGw1)	~\$60
Log Analytics	~\$10–20
Azure Comm. Services	\$0
Subtotal	~\$140–200

ACA containers scale to zero when idle. Real-world usage often 30–50% lower.

Total estimated cost: BBD \$140–240/mo — roughly half what a single ACA-only approach would cost, because the public side is absorbed by Cloudflare's free tier. The internal side's largest fixed cost is the VPN Gateway (\$60/mo).

4. Why This Split Makes Sense

Factor	Public (Cloudflare)	Internal (Azure)
Traffic pattern	Low, bursty, global	Few staff, occasional
Cost model	Free tier / pay-per-use	Pay-per-second + fixed VPN
Data sensitivity	Low (public info)	High (company confidential)
Auth required	None (rate limiting only)	Azure AD (existing Microsoft domain)
Network isolation	Edge-only	VNet + Private Endpoints
AI model	Workers AI (no server)	Self-hosted Ollama (full control)
Maintenance	Zero	Minimal (ACA auto-heals)
Staff access	N/A	Azure P2S VPN — staff use Microsoft login

The Cloudflare public side costs **\$0 in fixed infrastructure** — you only pay for AI compute (Workers AI, pennies per query). The Azure side provides the **isolation and control** needed for company data, while using the **Microsoft domain they already have** for authentication — no separate identity system needed.

5. Security & Compliance

Public (Cloudflare Edge)

- DDoS mitigation (any plan)
- WAF — OWASP Top 10, SQLi, XSS
- Full (Strict) SSL/TLS
- Rate limiting — 10 req/min per device
- Security headers — HSTS, XFO, CSP
- SOC 2, ISO 27001, PCI DSS

Internal (Azure VNet)

- Full network isolation (VNet)
- Private Endpoints — traffic never leaves Azure
- Azure AD authentication (VPN) — existing Microsoft accounts
- ACA managed identity — no connection strings
- Secrets as ACA secret references
- SOC 2, ISO 27001, HIPAA aligned

6. Microsoft Ecosystem Integration

Since the company already uses Microsoft (domain, Office 365, email), the internal architecture leverages that:

- **VPN authentication** — Azure VPN Gateway uses Azure AD. Staff authenticate with their existing Microsoft work credentials — no separate usernames/passwords
- **Azure Communication Services** — configured with the company's verified domain for sending emails as @supreme-counselling.com
- **Managed Identity** — ACA containers use Azure AD managed identities, no connection strings in code
- **Future-ready** — can easily add Azure OpenAI Service if Ollama is insufficient (same Azure AD auth)

Staff don't need to learn anything new — they log into the VPN client with the same Microsoft credentials they use for email. No additional identity infrastructure required.

7. OpenTofu Plan — Resources Provisioned

Module	Resources
Cloudflare Public	DNS records · Pages project · Pages Functions bindings (D1, R2, AI) · D1 database · R2 buckets · WAF rulesets · Rate limiting rules · SSL redirect · Security headers
Azure Internal	Resource group · Virtual Network · Subnets (ACA, PE, Gateway) · ACA environment (VNet-injected) · Container apps (RAG, Ollama) · Storage account · Azure Files shares · Private Endpoints · Private DNS zones · VPN Gateway (P2S, Azure AD auth) · Log Analytics

Code deployment: Cloudflare Pages + Functions are deployed via Wrangler CLI (or CI/CD). The infra plan creates all bindings — you just run `npm run wrangler pages deploy`.

8. Implementation Plan

Phase	Action	Effort
1	Configure Cloudflare account, verify domain	1 day
2	Run OpenTofu for Cloudflare module → Pages, D1, R2, WAF, DNS	1 hour
3	Deploy site via Wrangler: <code>npx wrangler pages deploy ./dist</code>	1 hour
4	Write Pages Functions (API endpoints using D1/R2/AI bindings)	1 day
5	Run OpenTofu for Azure module → VNet, ACA, Storage, VPN	2–3 hours
6	Configure Ollama on ACA + mount Azure Files	2 hours
7	Provision Azure Communication Services + configure email sending	1–2 hours
8	Staff onboarding — VPN client + Microsoft login	2 hours
9	End-to-end testing (bookings, chat, internal RAG)	1 day

Total estimated setup time: 3–5 days — most of this is writing the application code (Workers Functions). Infrastructure provisioning is under a day.

Appendix: Comparison to Original Azure-Only Design

	Original (Azure-only)	v2 (Cloudflare + Azure)
Public hosting cost	~BBD \$90–120/mo (ACA)	BBD \$0–40/mo (Cloudflare)
Internal cost	~BBD \$180–280/mo	~BBD \$140–200/mo
Total	~BBD \$270–400/mo	~BBD \$140–240/mo
AI inference	Self-hosted Ollama	Workers AI (public) + Ollama (internal)
Server management	Containers to maintain	Zero for public; minimal for internal
Staff auth	VPN certs or Azure AD	Azure AD (existing Microsoft accounts)
Data isolation	Separate storage	Separate storage + separate provider